

PRASAD'S PHYSICS

BRAHMAPUR

Assignment

Laws of motion and Friction

Numerical Problems

01. Find % increase in kinetic energy of a body if momentum increases by (i) 4% (ii) 150%. (iii) 200% (iv) 40%.
02. Find % decrease in kinetic energy of a body if momentum decreases by (i) 10% (ii) 60%. (iii) 30%
03. Find % increase in momentum of a body if kinetic energy increases by (i) 69% (ii) 96%.
04. Find % decrease in momentum of a body if kinetic energy decreases by (i) 51% (ii) 64%.
05. If kinetic energy of a body is doubled, what is the % increase in momentum
06. Find the acceleration of the block of 2kg mass when pulled by two forces, one along x-axis with 3N and other along y-axis with 4N.
07. A force of 15 N is applied on two blocks A & B of mass 1kg & 2kg. Find common acceleration & contact force between A & B.
08. A block of mass 'M' is attached to the lower end of a vertical rope of mass 'm'. An upward force 'p' acts on the upper end of the rope. The system is free to move. Find the force exerted by the rope on the block
09. A cricket ball of mass 100 gram moving with a velocity of 20 m/s is brought to rest by a player in 0.05 sec. Find the impulse of the force and average force applied by the player.
10. A car is moving at 40 km/h is to be stopped by applying brakes in a distance of 4m if mass of car is 2000 kg. What is the force required?
11. Find the horizontal force required to just move a body of mass 200 kg on a rough horizontal surface having co-efficient of friction 0.35
12. A horizontal force of 1.2 kg f is applied to a 1.5 kg block which rests on a horizontal surface. If the co-

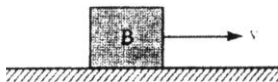
efficient of friction is 0.3, find the acceleration produced in the block. ($g = 9.8 \text{ ms}^{-2}$)

13. A body is pushed to give an initial velocity of 5 m/s. It comes to rest after traveling 5 m. Find coefficient of friction. ($g = 10 \text{ m/s}^2$)
14. A force 1000N is applied to a 1200 kg car if coefficient of friction is 0.04. What is the acceleration of the car ?
15. A motor car running at the rate of 7 ms^{-1} can be stopped by brakes in 10 m. Prove that the total resistance to the motor, when brakes are on, is one fourth of the weight of the car.

Multiple Choice Questions

01. When a body is stationary :
(a) There is no force acting on it.
(b) The combination of forces acting on it balances each other.
(c) The force acting on it are not in contact with it.
(d) the body is in vacuum.
02. A rocket is ejected 1kg of gases per sec at a velocity of 60 km/s. The accelerating force on the rocket is :
(a) 60 N (b) 600 N (c) 6000 N (d) 60,000 N
03. A block of metal weighing 2 kg is resting on a frictionless plane. It is stuck by a jet releasing water at a rate of 1kg/sec and a speed of 5 m/s. The initial acceleration of the block will be :
(a) 2.5 m/s^2 (b) 5 m/s^2 (c) 10 m/s^2 (d) 20 m/s^2
04. Two blocks of masses 2kg and 1kg are in contact with each other on a frictionless table. When a horizontal force of 3N is applied to the block of mass 2kg, the value of the force of contact between the two blocks is :
(a) 4N (b) 3N (c) 2N (d) 1 N
05. When we kick a stone, we get hurt, due to which one of the following properties of the stone does it happen?
(a) Inertia (b) Velocity
(c) Reaction (d) Momentum
06. A cricket player catches a ball of mass 10^{-1} kg , moving with a velocity of 25 m/s. If the ball is caught in 0.1s, the force of the blow exerted on the hand of the player is :

- (a) 4N (b) 25 N (c) 40 N (d) 250 N
07. You are marooned on a frictionless horizontal plane and cannot exert any horizontal force by pushing against the surface. How can you get off?
 (a) By jumping
 (b) By rolling your body on the surface
 (c) By splitting or sneezing
 (d) By running on the plane
08. A 5 kg stone falls from a height of 1000 m and penetrates 2m in a layer of sand. The time of penetration is :
 (a) 14.28s (b) 0.029 s (c) 7.14 s (d) 0.29 s
09. A sand bag of mass M is suspended by a rope. A bullet of mass m is fired at it with speed v and gets embedded in it. The velocity acquired by the bag is:
 (a) $mv/M + m$ (b) $Mv/M + m$
 (c) $v/M + m$ (d) mv/M
10. A bomb at rest explodes into two equal parts; one is moving due east with a velocity v then the other will:
 (a) Move due west with velocity v
 (b) come to rest
 (c) Move along north-west
 (d) Move along north-east
11. A man sitting in a train in motion is facing the engine. He tosses a coin up, the coin falls behind him. The train is :
 (a) Moving forward with acceleration.
 (b) Moving forward with uniform speed
 (c) Moving backward with uniform speed.
 (d) Moving backward with deceleration.
12. A block B is pushed momentarily along a horizontal surface with an initial velocity v. If μ is the coefficient of sliding friction between B and the surface, the block will come to rest after a time



- (a) $\frac{\mu g}{v}$ (b) $\frac{g}{v}$ (c) $\frac{v}{g}$ (d) $\frac{v}{\mu g}$
13. If the normal reaction is tripled, the force of limiting friction becomes.
 (a) three times (b) one-third
 (c) nine times (d) one-ninth
14. When one walks on the ground, the force of friction exerted by ground on him is in
 (a) forward direction
 (b) backward direction
 (c) first forward and then backward
 (d) zero.

15. The angle which the rough inclined plane makes with the horizontal when the body placed on it just starts sliding down is called
 (a) angle of friction. (b) angle of repose
 (c) critical angle (d) Brewster's angle
16. The limiting friction between two bodies in contact is independent of
 (a) nature of surfaces in contact
 (b) the area of surfaces in contact
 (c) normal reaction between the surfaces
 (d) all the above.
17. The frictional force is due to
 (a) Intermolecular attraction between the surfaces
 (b) Surface tension
 (c) repulsive forces
 (d) Interlocking between two bodies.
18. The maximum value of static friction is called
 (a) limiting friction
 (b) rolling friction
 (c) normal reaction
 (d) coefficient of static friction.
19. A block of mass 10 kg is placed on a rough horizontal surface having coefficient of friction $\mu = 0.5$. If a horizontal force of 100 N is applied on it, the acceleration of the block will be ($g = 10 \text{ m/s}^2$)
 (a) 15 m/s^2 (b) 10 m/s^2 (c) 5 m/s^2 (d) 0.5 m/s^2
20. A block of mass m is placed on a horizontal floor of a lift. If the lift begins to fall vertically with an acceleration $a < g$; then the frictional force on the block will be (μ_k = coefficient of kinetic friction, μ_s = coefficient of static friction)
 (a) zero (b) $\mu_k m (g - a)$
 (c) $\mu_s m (g + a)$ (d) $\mu_k ma$